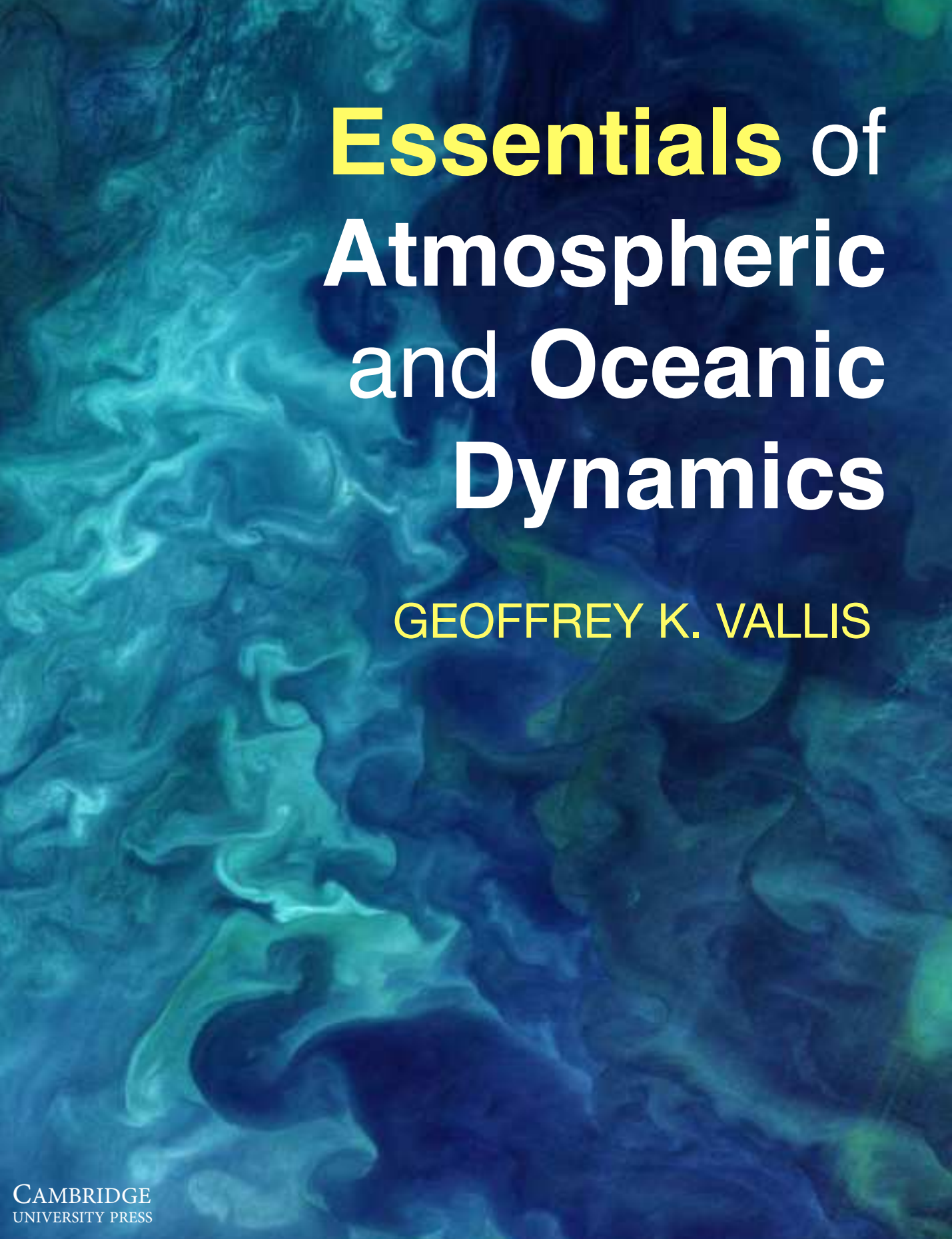




Essentials of **Atmospheric and Oceanic Dynamics**

GEOFFREY K. VALLIS

The "big book" [AOFD] by Vallis is a treasure, but I suspect that this new *Essentials* is destined to be used much more widely in classrooms. Vallis does a superb job of communicating the peculiar tensions between deductive reasoning and physical intuition that underlie this science. The new book is more approachable but no less rigorous. I especially appreciate how the various equation sets are derived in succinct but meaningful ways in the first few chapters, and then used as tools to explore the dynamics in the chapters that follow. It's almost the perfect introductory textbook on this subject, and I plan to use it in my own courses.

Brian E. J. Rose,
University at Albany

He's done it again. In *Essentials*, Geoff Vallis has produced a text that is useful to the student and the experienced scientist alike. While the content is simplified and shortened compared to its parent text, Vallis now provides even more descriptive explanations to support readers in their quest to navigate the physics of fluid flows. These explanations pair well with the theory, serving as an accessible introduction to students while also supporting the more experienced scientist as they put all of the pieces together. This will certainly be a future favourite for reading groups. Even readers with dog-eared versions of the parent book will want a copy of *Essentials*, for in it Vallis has added an entirely new chapter on planetary atmospheres, allowing the interested reader to venture into outer space to apply their newly honed GFD expertise.

Elizabeth A. Barnes,
Colorado State University

For the past decade, Geoff Vallis' book *Atmospheric and Oceanic Fluid Dynamics* has been the "go to" encyclopaedic resource, but it is too lengthy and comprehensive to use as a course textbook. With this superb new shorter volume, Geoff Vallis provides us with the definitive graduate-level textbook, with just the right balance of essential topics alongside glimpses of more advanced topics at the cutting edge of research. The extensive use of margin notes, diamonds to indicate advanced topics, and a comprehensive set of problems will ensure that *Essentials of Atmospheric and Oceanic Dynamics* has much to offer students and researchers at all levels. The book opens with the quote: "Seek simplicity, accept complexity. Exploit simplification, avoid complication." On all counts, this book succeeds magnificently!

David Marshall,
University of Oxford

Vallis' insights into the fundamentals and applications go a long way towards making otherwise complex topics readily grasped by those willing to study. He does not shy away from mathematics where needed, nor does he smother the reader with mathematics where pedagogically unnecessary. Those making it through this book will be ready to tackle a huge suite of research questions related to atmosphere and ocean fluid mechanics. Hence, this book serves an incredibly important role to the academic community. In a nutshell, we need more smart researchers who are adept at atmosphere and ocean dynamics to help understand how those dynamics are increasingly being affected by humanity's choices.

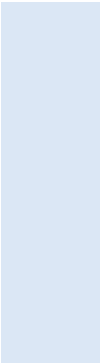
Essentials of Atmospheric and Oceanic Dynamics (EAOD) fills an important niche by offering an articulate and authoritative textbook to be worked through by advanced undergraduates and/or entering graduate students taking courses. The inclusion of exercises in EAOD is incredibly valuable for both students and teachers clamouring for more problem sets to test understanding. Whereas Vallis' previous book, *Atmospheric and Oceanic Fluid Dynamics* (AOFD) is the mother reference, EAOD offers a pedagogical entrée for those wishing to test the waters, including some deep waters. I will happily keep both books on my shelf and make use of them for personal study and to support the teaching of geophysical fluid dynamics.

Vallis has a clear writing style that brings the reader into the subject in an authoritative and friendly manner. He is a wise guru and gentle tutor. The subject of ocean and atmosphere fluid mechanics has matured greatly through his efforts at writing AOFD. EAOD furthers that maturation by allowing for a broader readership to tap into his brain. Well done Geoff!

Stephen M. Griffies,
GFDL, Princeton University.

As its parent book became the bible of the field, but also grew in size and the number of topics it covered in its latest edition, this new book provides a perfect balance and introduction to the essential topics, giving a quick reference without going into all the details. In the Vallis tradition, it is presented clearly, perfectly packaged, and is well organized for both atmospheric and oceanic fluid dynamics. Its simplicity will make it majestically appealing both for people outside the discipline looking for an accessible, yet complete, introduction, and for students within the field at all levels. The inclusion of planetary atmospheres broadens the scope and makes it appealing to a wider and growing audience. Anyone with a background in physics can get the essentials using this book.

Yohai Kaspi,
Weizmann Institute of Science



Contents

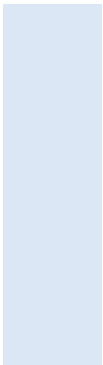
Preface	ix
 Part I GEOPHYSICAL FLUIDS	 1
1 Fluid Fundamentals	3
1.1 Time Derivatives for Fluids	3
1.2 The Mass Continuity Equation	7
1.3 The Momentum Equation	10
1.4 The Equation of State	12
1.5 Thermodynamics	13
1.6 Potential Temperature and Entropy	15
1.7 The Energy Budget	19
 2 Equations for a Rotating Planet	 24
2.1 Equations in a Rotating Frame of Reference	24
2.2 Equations of Motion in Spherical Coordinates	28
2.3 Cartesian Approximations: The Tangent Plane	32
2.4 Density Variations in the Atmosphere and Ocean	34
2.5 The Boussinesq Equations	36
2.6 Pressure Coordinates	40
 3 Dynamics on a Rotating Planet	 47
3.1 A Gentle Introduction to Scaling	47

3.2	Hydrostatic Balance	49
3.3	Geostrophic and Thermal Wind Balance	51
3.4	Static Instability and the Parcel Method	58
4	Shallow Water Equations	63
4.1	Shallow Water Equations of Motion	63
4.2	Conservation Properties	68
4.3	Shallow Water Waves	70
4.4	Geostrophic Adjustment	74
4.5	A Variational Perspective on Adjustment	79
5	Geostrophic Theory	83
5.1	Scaling the Shallow Water Equations	83
5.2	Geostrophic Shallow Water Equations	85
5.3	Scaling in the Continuously-Stratified System	89
5.4	Planetary-Geostrophic Equations for Stratified Fluids	92
5.5	Quasi-Geostrophic Equations for Stratified Fluids	93
5.6	The Two-Level Quasi-Geostrophic Equations	96
5.7	Frictional Geostrophic Balance and Ekman Layers	99
6	Rossby Waves	104
6.1	Fundamentals and Formalities	104
6.2	Group Velocity	108
6.3	Rossby Wave Essentials	109
6.4	Rossby Waves in Stratified Quasi-Geostrophic Flow	114
6.5	Ray Theory and Rossby Rays	118
	Appendix A: The WKB Approximation for Linear Waves	119
7	Gravity Waves	123
7.1	Internal Waves in a Continuously-Stratified Fluid	123
7.2	Properties of Internal Waves	125
7.3	Internal Waves in a Rotating Frame of Reference	132
7.4	Topographic Generation of Internal Waves	135
7.5	Flow over an Isolated Ridge	140
8	Instability	144
8.1	Kelvin–Helmholtz Instability	144
8.2	Instability of Parallel Shear Flow	146
8.3	Necessary Conditions for Instability	150
8.4	Baroclinic Instability	151
8.5	The Eady Problem	155

8.6	Two-Level Baroclinic Instability	161
8.7	Necessary Conditions for Baroclinic Instability	167
9	Waves and Mean-Flows	170
9.1	Quasi-Geostrophic Wave–Mean-Flow Interaction	170
9.2	Potential Vorticity Flux	174
9.3	The Transformed Eulerian Mean	178
9.4	The Non-Acceleration Result	182
10	Turbulence	188
10.1	The Problem of Turbulence	188
10.2	Turbulent Diffusion	191
10.3	Spectral Theories of Turbulence	193
10.4	Two-Dimensional Turbulence	198
10.5	Geostrophic Turbulence	203
10.6	Beta-Plane Turbulence	208
Part II	ATMOSPHERES	213
11	The Tropical Atmosphere	215
11.1	An Observational Overview	215
11.2	An Ideal Hadley Circulation	217
11.3	Instability of the Ideal Hadley Circulation	221
11.4	The Tropical Circulation	222
11.5	Effects of Moisture	228
11.6	Tropical Convection	232
12	Midlatitudes and the Stratosphere	237
12.1	Jet Formation and Surface Winds	237
12.2	The Ferrel Cell	245
12.3	The Residual Ferrel Cell	249
12.4	Stratosphere	251
13	Planetary Atmospheres	257
13.1	A Taxonomy of Planets	258
13.2	Dimensional and Nondimensional Parameters	260
13.3	Radiative Transfer and Vertical Structure	269
13.4	Planets in our Solar System	273
13.5	Exoplanets and Tidal Locking	283

Part III OCEANS	287
14 Wind-Driven Gyres	289
14.1 An Observational Overview	289
14.2 Sverdrup Balance	290
14.3 Ocean Gyres	291
14.4 Effects of Nonlinearity	300
14.5 An Inertial Solution	302
15 The Overturning Circulation and Thermocline	305
15.1 The Observations	305
15.2 A Mixing-Driven Overturning Circulation	307
15.3 Thermocline Dynamics and the MOC	310
15.4 An Interhemispheric Overturning Circulation	315
15.5 The Antarctic Circumpolar Current	320
16 Equatorial Oceans and El Niño	324
16.1 Observations of the Equatorial Ocean	324
16.2 Vertically Integrated Flow and Sverdrup Balance	325
16.3 Dynamics of the Equatorial Undercurrent	328
16.4 El Niño and the Southern Oscillation	333
16.5 Ocean Dynamics in El Niño	339
16.6 Unstable Air–Sea Interactions	342
Bibliography	346
Index	352

Note: In the text itself more advanced sections are marked with a diamond, ♦, and may be omitted on a first reading. If a section is so marked then the marking applies to all the subsections within it.



Preface

*Seek simplicity, accept complexity.
Exploit simplification, avoid complication.*

This is an introductory book on the dynamics of atmospheres and oceans, with a healthy dose of geophysical fluid dynamics. It is written roughly at the level of advanced or upper-division undergraduates and beginning graduate students, but parts of it will be accessible to first- or second-year undergraduates and I hope that practising scientists will also find it useful. The book is designed for students and scientists who want an introduction to the subject but who may not want all the detail, at least not yet, and its prerequisites are just familiarity with some vector calculus and basic classical physics. Thus, it is meant to be accessible to non-specialists and students who will not necessarily go on to become professional dynamists. However, as well as very basic material the book does include some elementary introductions to a few ‘advanced’ topics, such as the residual circulation and turbulence theory, as well as material on the general circulation of the atmosphere and ocean. The more advanced parts could easily be omitted for a first course and, like difficult ski slopes, are marked with a diamond, ♦. Readers may explore these topics more in the references provided, or in this book’s parent, *Atmospheric and Oceanic Fluid Dynamics*. Nearly all the topics in this book, except those in the chapter on planetary atmospheres, are dealt with in greater detail there.

What is in the book

The book is divided into three Parts. The first, and longest, provides the foundation for the study of the dynamics of the atmosphere and ocean. It does not assume any prior knowledge of fluid dynamics or thermodynamics, although readers who have such knowledge may be able to skim Chapter 1. The rest of Part I provides an introduction to ‘geophysical fluid dynamics’, the subject that remains at the heart of atmospheric and oceanic dynamics and without which the subject would be largely qualitative and/or computational. Here we discuss the effects of rota-

tion and stratification, leading into shallow water theory and the quasi-geostrophic and planetary-geostrophic equations. Rossby waves, gravity waves, baroclinic instability and elementary treatments of wave–mean-flow interaction and turbulence round out Part I.

Parts II and III focus on the large scale dynamics and circulation of the atmosphere and ocean, respectively. Our main focus in both Parts is what is sometimes called ‘the general circulation’, meaning the large-scale quasi-steady and/or time-averaged circulation, but this circulation depends on the effects of time-dependent eddies — the atmosphere’s Ferrel Cell may be considered to be ‘driven’ by the effects of baroclinic instability and Rossby waves. And the El Niño phenomenon, described in the final chapter, is explicitly time dependent. One feature of this book that is not in the parent book is a chapter discussing some of the general principles of planetary atmospheres, a topic of increasing interest because of the new, sometimes quite spectacular, observations of the planets in our Solar System and beyond.

How to use the book

The contents of the book are about enough for a two-term course in atmosphere–ocean dynamics. A term-long, first course in geophysical fluid dynamics could, for example, be based on Part I, omitting some of the earlier or later chapters depending on the students’ backgrounds and interests. A term-long course in atmospheric and/or oceanic circulation could be based on Part II and/or Part III, supplementing the material with review articles or research papers as needed, perhaps using data sets to look at the real world (and other planets, if Chapter 13 is to be studied). Alternatively, one could combine aspects of Parts I and II, or Parts I and III, to construct an ‘Atmospheric Dynamics’ or ‘Oceanic Dynamics’ course.

If the book is to be used for self-study it could simply be read from beginning to end, although many other pathways are possible and may be preferable. Parts II and III depend on the material in Part I, but the material is reasonably self-contained, and readers who already have some knowledge of geophysical fluid dynamics should feel free to start at a later chapter, or with Part II or Part III. A few problems are collected at the end of some chapters; these are designed to test understanding as well as to fill in gaps and extend the material in the book itself. Many other problems at varying levels of difficulty can be found on the web site of this book, which can easily be found with a search engine. The reader will also see a number of margin notes throughout the book, rather like the ones to the left. The book itself was typeset using $\text{\LaTeX}$ with Crimson fonts for text, Cronos Pro for sans serif and Minion Math for equations.

I would like to thank Matt Lloyd, Zoë Pruce and Richard Smith at Cambridge University Press for their expert guidance through the writing and production process, as well as many colleagues and students — too many to list, but they know who they are — for their many comments, corrections and criticisms. If you, the reader, have other comments, major or minor, do please contact me.

Margin notes that are set in a roman (i.e., upright) font emphasize or expand on something that is in the main text.

Margin notes set in italics are asides or historical anecdotes.

Part I

GEOPHYSICAL FLUIDS

Fluid Fundamentals

F LUIDS, LIKE SOLIDS, move if they are pushed and they warm if they are heated. But, unlike solids, they flow and deform. In this chapter we establish the governing equations of motion for a fluid, with particular attention to air and seawater — the fluids of the atmosphere and ocean, respectively. Readers who already have knowledge of fluid dynamics may skim this chapter and begin reading more seriously at Chapter 2, where we begin to look at the effects of rotation and stratification.

1.1 TIME DERIVATIVES FOR FLUIDS

1.1.1 Field and Material Viewpoints

In solid-body mechanics one is normally concerned with the position and momentum of an identifiable object, such as a football or a planet, as it moves through space. In principle we could treat fluids the same way and try to follow the properties of individual fluid parcels as they flow along, perhaps getting hotter or colder as they move. This perspective is known as the *material* or *Lagrangian* viewpoint. However, in fluid dynamical problems we generally would like to know what the values of velocity, density and so on are at *fixed points* in space as time passes. A weather forecast we care about tells us how warm it will be where we live and, if we are given that, we may not care where a particular fluid parcel comes from or where it subsequently goes. Since the fluid is a continuum, this knowledge is equivalent to knowing how the fields of the dynamical variables evolve in space and time. This viewpoint is known as the *field* or *Eulerian* viewpoint.

Although the field viewpoint will often turn out to be the most practically useful, the material description is invaluable both in deriving the equations and in the subsequent insight it frequently provides. This is because the important quantities from a fundamental point of view are

The fluid dynamical equations of motion determine the evolution of a fluid. The equations are based on Newton's laws of motion and the laws of thermodynamics, and embody the principles of conservation of momentum, energy and mass. Initial conditions and boundary conditions are needed to solve the equations.

The Lagrangian viewpoint is named for the Franco-Italian J. L. Lagrange (1736–1813), one of the most renowned mathematicians of his time. The Eulerian point of view is named for Leonhard Euler (1707–1783), the great Swiss mathematician. In fact, Euler is also largely responsible for the Lagrangian view, but the attribution became tangled over time.

often those which are associated with a given fluid element: it is these which directly enter Newton's laws of motion and the thermodynamic equations. It is thus important to have a relationship between the rate of change of quantities associated with a given fluid element and the local rate of change of a field. The material derivative (also called the advective derivative or Lagrangian derivative) provides this relationship.

1.1.2 The Material Derivative of a Fluid Property

A *fluid element* is an infinitesimal, indivisible, piece of fluid — effectively a very small fluid parcel of fixed mass. The *material derivative*, or the *Lagrangian derivative*, is the rate of change of a property (such as temperature or momentum) of a particular fluid element or finite mass of fluid; that is, it is the total time derivative of a property of a piece of fluid.

Let us suppose that a fluid is characterized by a given velocity field $\mathbf{v}(\mathbf{x}, t)$, which determines its velocity throughout. Let us also suppose that the fluid has another property φ , and let us seek an expression for the rate of change of φ of a fluid element. Since φ is changing in time and in space we use the chain rule,

$$\delta\varphi = \frac{\partial\varphi}{\partial t}\delta t + \frac{\partial\varphi}{\partial x}\delta x + \frac{\partial\varphi}{\partial y}\delta y + \frac{\partial\varphi}{\partial z}\delta z = \frac{\partial\varphi}{\partial t}\delta t + \delta\mathbf{x} \cdot \nabla\varphi. \quad (1.1)$$

This is true in general for any $\delta t, \delta x$, etc. The total time derivative is then

$$\frac{d\varphi}{dt} = \frac{\partial\varphi}{\partial t} + \frac{d\mathbf{x}}{dt} \cdot \nabla\varphi. \quad (1.2)$$

If this equation is to provide a material derivative we must identify the time derivative in the second term on the right-hand side with the rate of change of position of a fluid element, namely its velocity. Hence, the material derivative of the property φ is

$$\frac{d\varphi}{dt} = \frac{\partial\varphi}{\partial t} + \mathbf{v} \cdot \nabla\varphi. \quad (1.3)$$

The right-hand side expresses the material derivative in terms of the local rate of change of φ plus a contribution arising from the spatial variation of φ , experienced only as the fluid parcel moves. Because the material derivative is so common, and to distinguish it from other derivatives, we denote it by the operator D/Dt . Thus, the material derivative of the field φ is

$$\frac{D\varphi}{Dt} = \frac{\partial\varphi}{\partial t} + (\mathbf{v} \cdot \nabla)\varphi. \quad (1.4)$$

The brackets in the last term of this equation are helpful in reminding us that $(\mathbf{v} \cdot \nabla)$ is an operator acting on φ . The operator $\partial/\partial t + (\mathbf{v} \cdot \nabla)$ is the *Eulerian representation of the Lagrangian derivative as applied to a field*.